

# Appendix 3.4: Regulations on Age Capacity, Local OCR Compliance, and Adaptive Biometrics

### Document Metadata

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# 1. Age Graduation and Institutional Sovereignty

This regulation defines the operational parameters for user legal capacity, client-side compliance verification, and adaptive biometric anchoring within the Transceptor Global institutional contour. System architecture maintains absolute sovereign precedence over execution parameters, enforcing age-based operational constraints directly via client-side cryptographic attestations.

### Operational Legal Capacity Tiers

- **Tier 0: Restricted Custody (< 14 Years)**
  - Operations are strictly restricted or executed under an explicit, cryptographically signed custodial mandate.
  - Direct execution of un-escorted financial transactions is prohibited.
- **Tier 1: Junior Capacity (14–17 Years)**
  - Limited operational capacity with strict velocity and value caps.
  - Requires guardian co-signing via threshold cryptographic signatures for transactions exceeding defined baseline risk parameters.
- **Tier 2: Full Legal Capacity (18+ Years)**

    - Unrestricted transactional access across all network contours.
    - Complete operational sovereignty and authorization privileges subject to standard AML/CFT risk scoring.

### Institutional Sovereignty Paradigm

Local jurisdictional requirements are reconciled through zero-knowledge proofs (ZKPs) generated locally within client devices. Institutional sovereignty ensures that raw Personally Identifiable Information (PII) is never transmitted to or stored on central servers, maintaining full user privacy while guaranteeing immutable compliance proofs across distributed nodes.

# 2. Local Passport OCR Parsing and 1-to-1 Face Match in TEE

To ensure absolute compliance without violating privacy constraints, document validation, Optical Character Recognition (OCR), and biometric face matching are executed entirely inside the local device's Hardware-isolated Trusted Execution Environment (TEE), such as Apple Secure Enclave or ARM TrustZone.

### Local Processing & Verification Protocol

1. **Document Ingestion:** The client device extracts the visual inspection zone (VIZ) and Machine Readable Zone (MRZ) of legal identification documents.
2. **Local TEE Execution:** Raw document images are parsed locally. No unencrypted PII data leaves the hardware boundary.
3. **1-to-1 Facial Matching:** The 512-dimensional facial embedding vector derived from the document photo $V_{doc}$ is compared against a live, liveness-verified facial vector $V_{live}$.

### TEE OCR & Biometric Matching Pipeline

| Stage | Process Name | Data Inputs |
| --- | --- | --- |
| **01** | Capture & Frame Quality Validation | Raw Video Feed, Depth Sensors |
| **02** | Local OCR & MRZ Parsing | Normalized Document Frame |

| Stage | Process Name | Data Inputs |
| --- | --- | --- |
| **03** | Liveness & Feature Vectorization | Front Camera Stream, Sensor Fusion |
| **04** | 1-to-1 Vector Comparison | $V_{live}$ and Document Vector $V_{doc}$ |
| **05** | ZK-Proof Generation | Match Flag, Age Threshold |

# 3. Adaptive Biometric Anchor

Human physical features naturally change over time due to aging, injury, or stylistic changes. To prevent identity lock-outs without compromising security, the system maintains a dynamically updated Adaptive Biometric Anchor.

## Dynamic Vector Update Formulation

Upon successful multi-factor authentication and high-confidence liveness verification, the static reference biometric anchor vector $V_{anchor}$ is updated incrementally:

$$V_{anchor}^{(t+1)} = (1 - \alpha) \cdot V_{anchor}^{(t)} + \alpha \cdot V_{verified}^{(t+1)}$$

Where:

- $V_{anchor}^{(t)}$ represents the current biometric anchor vector stored in secure device storage.
- $V_{verified}^{(t+1)}$ represents the freshly captured biometric vector from a verified session.
- $\alpha \in (0, 0.05]$ is the adaptive learning rate parameter governing update velocity to prevent adversarial drift.

## Update Qualification Conditions

An anchor update is executed if and only if the following criteria are simultaneously satisfied:

- **Liveness Confidence:** $C_{live} \ge \theta_{liveness}$ (where $\theta_{liveness} = 0.99$).

- **Match Similarity:** $S(V_{verified}, V_{anchor}) \ge \theta_{match}$ (where $\theta_{match}$ represents a high-confidence similarity threshold).
- **Environment Integrity:** Device attestation confirms execution in an uncompromised hardware environment.

# 4. Cascade Recovery Protocol for Appearance Changes

When dynamic biometric drift exceeds acceptable mathematical bounds (e.g., radical facial surgical alteration, severe trauma, or multi-year inactivity), the system triggers the automated **Cascade Recovery Protocol**.

## Recovery Level Escalation Pipeline

1. **Level 1: Multi-Angle Dynamic Liveness Retry**
    - Initiates multi-axis head rotation, depth mapping, and variable lighting response testing to isolate transient visual anomalies.
2. **Level 2: Hardware-Backed Fallback (FIDO2 / WebAuthn)**
    - Demands primary authorization via a physical WebAuthn/FIDO2 hardware key combined with the primary secure device PIN/passphrase.
3. **Level 3: Cryptographic Guardian Social Recovery**
    - Triggers a $k$-of-$n$ threshold signature recovery protocol managed by pre-designated trusted guardians or secondary secure nodes.
4. **Level 4: In-TEE Re-Verification & Re-Anchoring**
    - Prompts a complete re-verification cycle using official government identity documentation parsed locally via TEE OCR, replacing $V_{anchor}$ with a newly established baseline.